the reliability and coverage of the first 99 years of the data are unclear, and we opted to exclude them from the analysis.

The consensus partition, obtained by collecting the outcomes of 100 runs of the hSBM algorithm, consists of 7 hierarchical levels. On the left of Fig. 1(c), we draw this hierarchy only until the 3rd level (starting from the common root at level 0) for visualization clarity. More information on the number of clusters in each level can be found in Sec. S2.1 in the SM, where we also conduct a comparative analysis of the consensus partition with the individual runs of the hSBM algorithm.

On the right of Fig. 1(c), we also show the heatmap of the number of papers in time for each cluster at the 3rd level, for a total of 16 different clusters after excluding other 5 clusters with less than 500 papers not included in this analysis. The keywords shown in the heatmap have been manually assigned to roughly characterize each cluster. In particular, the keywords between parentheses have been chosen amongst the most frequent and most significant in the clusters at the 4th level, while the most representative keyword at the 3rd level has been chosen and printed in bold. For details on how they were assigned see *The pipeline* section and Sec. S2.2 in the SM.

In the following, we refer to a cluster at the 3rd level by its representative keyword (capitalized). As shown by the hierarchy and by the horizontal red lines in the heatmap, clusters are divided in three main branches. Looking at the two biggest branches, we can see a clear division between more STEM oriented documents (top branch) and those in Political sciences, Social sciences, as well as Medicine (bottom branch). Notably, a third smaller branch appears isolated, including papers at the intersection of the other two, mostly about Wireless technologies and their applications. Going into more details, in the STEM branch we notice how Cybersecurity, Control theory and Telecommunication (clusters 1, 6, and 10 respectively) are the ones producing most publications, with Blockchain (cluster 3) becoming the most relevant in the last 5 years in terms of number of papers published per year. On the other branch, Governance (cluster 12), including works in Political science, Education and Fiscal federalism, is the most relevant cluster, while Environment, Social network analysis and Health clusters (respectively clusters 13, 14, and 15) have produced a smaller number of papers. Furthermore, see Fig. S5 and Fig. S6 in the SM respectively for a similar plot done at the 4th level and for a bipartite hierarchical network showing how clusters are represented in the various topics.

In Fig. 1(c) we have shown how the number of papers in each of these clusters has evolved over time. Looking more into details of the early history of *(de)centralization*, the first papers adopting the term have all been in the Social sciences branch, most importantly the Governance cluster, followed by Social network analysis and Health. In the 1950s, indeed, there are 58 papers in Governance, which represents the first cluster to adopt the term and use it extensively. Some of these articles refer, among other things, to democracy as a form of centralized decision-making system [26]. Other clusters with more than 10 papers refer to Social network analysis and Health, as seen for example in Kaufman's "Toward an interactional conception of community" [27]. Here, *centralization* is depicted as a force gradually destroying the concept of community as a social unit. Notably, most of these papers have no citations from other articles in the full corpus, with only some citations within the cluster of governance. In the 1960s, the largest growth is found again in the Governance and Health clusters, both reaching around 150 papers each in the decade. An important example of the former is that of Bachrach et al. [28], where they highlight how different disciplines (i.e., social and political sciences) reach completely opposite conclusions about the *(de)centralization* of power. In the same decade *(de)centralization* also appears in other relevant clusters, namely Social network analysis (50 papers) and Investments (29 papers), with a

significant number of citations in both directions between them. The term is picked up from the STEM branch only later in the 1970s, especially through works in Control theory and Optimization [29], coming significantly to a popular domain as Cybersecurity only in the 1980s.

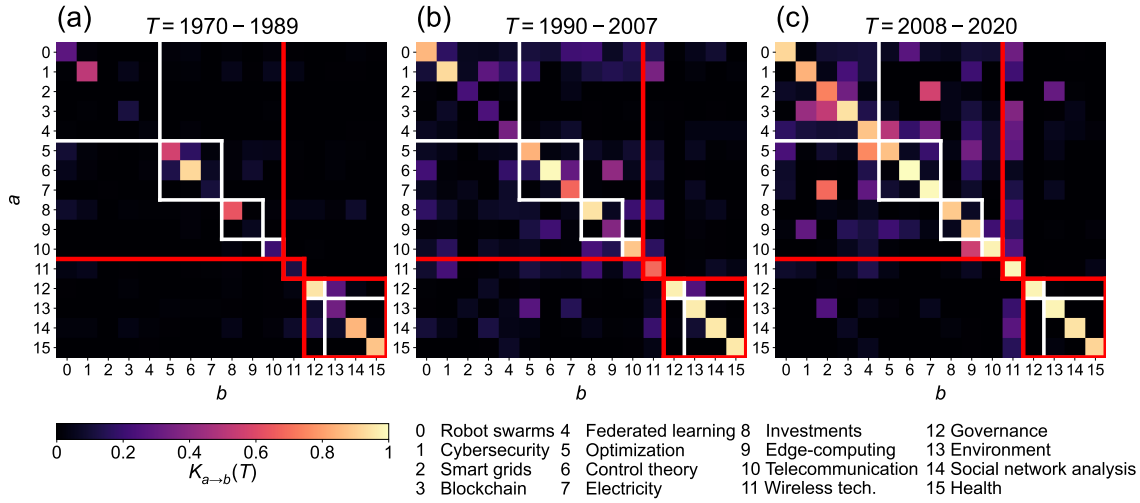

Figure 2: **Temporal evolution of the influence between clusters.** Average knowledge flows $K_{a\to b}(T)$ from each cluster a to each cluster b at level 3 in the period T = 1970-1989 **(a)**, T = 1990-2007 **(b)** and T = 2008-2020 **(c)**, represented by the different colors according to the colorbar in the bottom left part of the figure. A representative keyword for each cluster is reported on the bottom right part of the figure. White lines denote clusters belonging to the same 2nd level cluster, whereas red lines mark different branches at the 1st level. In the first period, little to no communication is happening between different clusters. In recent years, more communication happens inside the same 2nd level cluster, and towards the middle branch (cluster 11). However, little communication happens between the two other branches, roughly representing the STEM and social sciences communities respectively.

We have seen how different domains have picked up the concept of *(de)centralization* at different times. It is therefore natural to ask whether they developed such uses separately, or they influenced each other in some way. The hierarchical clustering partially answers this question, as it gives a degree of separation between domains based on citation and semantic information. However, significant information is still present in the citations between papers of different clusters. We exploit this by computing knowledge flows [23], whose aim is to quantify the transfer of knowledge given by the citations between groups of papers through a comparison with a null model (for more details see *The pipeline* Section). We thus study the average knowledge flow $K_{a\to b}(T)$ from papers in a cluster a, at level 3 of the hierarchy, in a period of time T to future publications in another cluster b, represented by a number between 0 and 1 showing how significant this influence has been.

Here, in Fig. 2 we consider three different periods of time T: 1970-1989, 1990-2007 and 2008-2020. Similarly to what we will do in the next figures, we have excluded the year 2021 as a source of knowledge flow, because our dataset ends at the end of 2021, thus meaning that we cannot evaluate knowledge flows from papers of 2021 to future years. In the figure, all clusters are ordered as in Fig. 1(c), with the representative keywords shown in the legend below. For each period T, the color of the cell of row a and column b of the heatmap refers to the average knowledge flow $K_{a\to b}(T)$ from papers in cluster a in that period of time to future papers in cluster b, according to the colormap shown below.

Starting from the first period in Fig. 2(a), between 1970 and 1989 clusters have little to no influence on the future of the other ones. We have previously seen how in these years the use of *(de)central-*

ization started to rise across some domains, mostly being Governance, Control theory, Social network analysis, Health, Cybersecurity, and Investments. However, apart from Governance and Control theory (clusters 12 and 6), these clusters have low knowledge flow even to themselves, meaning that the use of *(de)centralization* was only relegated to sporadic and not so influential papers in the literature. This also confirms that the topic has appeared independently at this early stage.

Then, looking at the period from 1990 and 2007 in Fig. 2(b), we can see how much more transfer of knowledge has occurred between clusters. As shown in Fig. 2(c), this trend is even more pronounced in the last and more recent period, whose start coincides with the creation and rise of blockchain technologies. Interestingly, these transfers reflect the structure of the hierarchy and denote significant differences between the high-level domains. The STEM branch (made of the clusters 0 to 10) shows clear communication between clusters belonging to the same group both at the 2nd and 1st level (respectively within white and red lines), whereas the right bottom branch shows almost no communication with the other domains, especially after 2008. The only significant knowledge flow from this branch in the middle period goes from Environment (cluster 13) to Optimization (cluster 5), while in the last period this is only relegated between Environment and Smart grids (cluster 2). The middle branch instead shows clear influence from the other two, and little influence towards them, especially in the last period.

Finally, notice how the highest knowledge flows between different clusters in Fig. 2(b) are from those that, in the first period, were starting to be more influential. Similarly in the last period the clusters with highest influence on other ones are mostly within the STEM branch. Overall, the heatmaps show a clear decentralized birth of the concept of *(de)centralization*, appearing in different fields and domains with little to no communication between each other. Instead, in recent years, we find a more coordinated evolution, even though still sectorial in some cases, and mainly lead by STEM related clusters.

3.2 The case of Governance and Blockchain

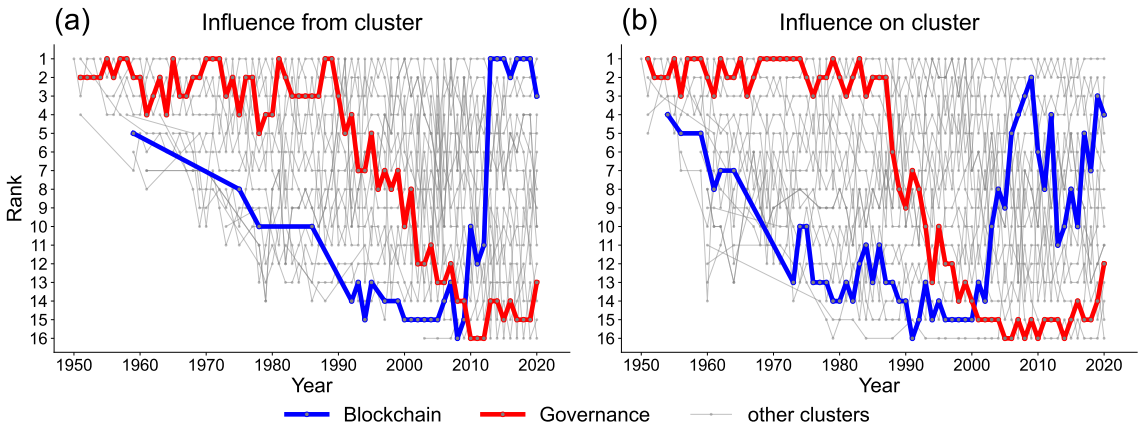

Figure 3: **Ranking the influence of Blockchain and Governance in the *(de)centralization* literature. (a)** Ranking in time of the influence coming from a cluster in the 3rd hierarchical level, computed on the average knowledge flow $K_{a\to\bullet}(Y)$ from papers in cluster a published in the year Y towards all other future papers. **(b)** Ranking in time of the influence to a cluster in the 3rd hierarchical level, computed on the average knowledge flow $K_{\bullet\to a}(Y)$ from papers published in the year Y towards future papers in cluster a. The blockchain cluster, highlighted in blue, has become a central actor in the recent literature on *(de)centralization*, supplanting the governance cluster, highlighted in red.

Having analyzed the concept of *(de)centralization* in the general academic landscape, we now focus